

ULTRON 2.7 Phase 2 Execution Report

Safe low-risk laptop execution layer

Project	ULTRON 2.7
Phase	Phase 2: Safe Low-Risk Execution
Workspace	Local workspace checkout: U2.7
Status	Implemented and verified

Executive Summary

Phase 2 upgrades ULTRON from a dry-run-only prototype into a controlled laptop assistant that can perform selected low-risk actions for real.

The implementation adds safe-root file boundaries, application aliases, real note/reminder/timer persistence, safe file search/opening, clipboard helpers, screenshot support, and stronger Windows console output handling.

Phase 2 Scope

- Keep the Phase 1 planner, validator, policy, and audit flow intact.
- Allow real execution only for low-risk tools.
- Restrict file and folder tools to configured safe roots.
- Add config fields for app aliases, safe roots, and screenshot output.
- Keep destructive actions non-destructive even after confirmation.
- Add tests for safe-root behavior and real low-risk writes.

What Was Implemented

Area	File(s)	Result
Configuration	src/ultron27/config.py	Added safe_roots, app_aliases, and screenshot_dir.
Executor	src/ultron27/executor.py	Added real handlers for low-risk laptop actions.
Planner	src/ultron27/planner.py	Added regex plans for notes, reminders, timers, screenshot, clipboard, and folders.
Tool schema	src/ultron27/tools.py	Added clipboard tool schemas.
CLI	src/ultron27/cli.py	Passes config to executor and emits Windows-safe JSON.

Area	File(s)	Result
Docs	README.md, docs/architecture.md, docs/phase2_execution.md	Documents Phase 2 behavior and guardrails.
Tests	tests/test_pipeline.py	Expanded to 15 regression tests.

Safety Boundary

Category	Tools	Behavior
Real low-risk	notes, reminders, timers, search/open safe files, app launch	Allowed after validation and policy allow.
Optional low-risk	screenshot, clipboard	Runs only when OS/library support exists.
Protected high-risk	delete file, send email, run script, shutdown, restart	Policy may confirm, but executor remains not_implemented.
File boundary	search/open file/folder	Limited to configured safe_roots.

Verification Results

Check	Command	Result
Unit tests	python -m unittest discover -s tests	15 tests passed.
Dataset evaluation	python scripts/evaluate_dataset.py --split test	Intent 1.0, tool 1.0, argument exact match 0.9614, confirmation 1.0.
Screenshot dry-run	python -m ultron27 "take a screenshot" --no-audit	Valid low-risk dry-run response.
Real note write	python -m ultron27 "create note phase two" --execute --workspace . --no-audit	Created notes/phase_two.md.
Confirmed delete	python -m ultron27 "delete project_report.txt" --yes --execute --no-audit	Returned not_implemented, so no destructive action occurred.

Recommended Next Step

Phase 3 should focus on dataset quality and evaluation: deduplicate conflicting synthetic rows, add real usage logs, add Hinglish/Hindi/STT-error variants, and create a separate safety regression split.